Curriculum Vitae

Personal Information

Name: WU, Jingxian
Position Sought: Intern or Co-op
Tel: (573) 884-0097
Date of Birth: November 25, 1975
Major: Electrical and Computer Engineering
E-Mail: jwu@ee.missouri.edu

Education

2001 ~ Present **University of Missouri – Columbia** **MO, USA**
- *Candidate of Ph.D.,* Expected: May, 2005
- *Grade Average*: 4.0/4.0

1998 ~ 2001 **Tsinghua University** **Beijing, P. R. China**
- *Master of Science,* Jan., 2001
- *Rank*: top 3 out of 72

1994 ~ 1998 **Beijing University of Aero. & Astro.** **Beijing, P. R. China**
- *Bachelor of Engineering,* July, 1998
- *Rank*: top 2 out of 112

Research Experiences

2001 ~ Present **The Digital Communication Lab** **MU**

Research Assistant

- Joint turbo decoding and channel equalization for systems over doubly selective fading channels
- Theoretical performance analysis of diversity systems over fading channels
- Optimum diversity combining with non-ideal channel information
- Discrete-time channel modeling for MIMO wireless communication systems
- Multi-user channel estimation for broadband CDMA communication systems

May 2003-Aug. 2003 **Mitsubishi Electric Research Lab**

Internship Engineer

- Transmitter algorithm design for 3GPP high speed downlink packet access (HSDPA) system

1999 ~ 2001 **The Broadband Access Network Lab** **Tsinghua University**

Research Assistant

- Build the network layer simulation modules for hybrid fiber coaxial (HFC) network with C++ and the Network Simulator *ns*-2
- HFC return path noise suppression
- Developed the embedded operational software for Cable Modem, on the platform of PSOS
- Designed and implemented the hardware architecture of Cable Modem

1998 ~ 1999 **The Tsinghua–Motorola Personal Communication Lab** **Tsinghua University**

Research Assistant

- Designed a dictionary engine for Personal Digital Assistant (PDA).
- A developing member of Motorola's PDA product MobilePAD828™

- A developing member of Motorola's mobile phone product A6188™

1997 ~ 1998 **The Synthetic Aperture Radar Research Lab** **BUAA**

Research Assistant as an undergraduate student

- Independently designed and developed the software of VisualSAR, a visualized and integrated SAR image analysis system on the UNIX platform

1997 **The Robot Research Institute** **BUAA**

Research Assistant as an undergraduate student

- Designed and implemented an interface circuit for a high-building-window-cleaning Robot.

Professional Activities

- Internship engineer in Mitsubishi Electric Research Lab (MERL) (2003)
- Internship engineer in Beijing I-Star Company (1999)
- Reviewer for IEEE Journal on Selected Areas of Communications
- Reviewer for IEEE Transactions on Wireless Communications
- Reviewer for IEEE Transactions on Transaction on Vehicular Technology
- Reviewer for IEEE Transactions on Circuits and Systems - I
- Reviewer for International Journal on Wireless Information Networks
- Member, Tau Beta Pi, Sigma Xi
- Student member, IEEE

Academic Honors

- Research Assistantship (2001 ~ 2004 MU)
- Guanghua Fellowship (1999, Tsinghua University**)**
- Internship Scholarship, First Prize (1999, Tsinghua University)
- Privilege to be enrolled in the master program of Tsinghua University with entrance examination waived (1998, Tsinghua University)
- The People's Scholarship, First Prize (1995, 1996, 1997, BUAA)
- Second Awards of the Advanced Mathematical Competition (1996, BUAA)
- Lizheng Fellowship for the top 1 freshman (1994, BUAA)

Skills

- Proficient with computer programming in C/C++ on both Windows and UNIX platform
- Communication systems design and simulation with Matlab
- Network layer simulation for communication networks
- Software design and hardware implementation of communication systems
- Familiar with PCB design tools, such as PowerPCB, Protel, ORCAD
- Linux system administration

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Publications

Journal Papers

1. **Jingxian Wu**, Chengshan Xiao, and Norman C. Beaulieu, "Optimal Diversity Combining with Noisy Channel Estimation", submitted to *IEEE Trans. Commun.*, 24 pages, 2004.
2. Sang-Yick Leong, **Jingxian Wu**, Jan C. Olivier, and Chengshan Xiao, "Fast Time-varying Dispersive Channel Estimation and Equalization for 8-PSK Cellular System", ***revised version*** submitted to *IEEE Trans. Veh. Technol.*, 25 pages, 2003
3. **Jingxian Wu** and Chengshan Xiao, "Multiuser channel estimation for CDMA systems over Doubly Selective fading channels", ***revised version*** submitted to *IEEE Trans. Wireless Commun.*, 26 pages, 2003.
4. **Jingxian Wu**, Chengshan Xiao, and Jan C. Olivier, "Time-Varying and frequency-selective channel estimation with unequally spaced pilot symbols", submitted to *International J. Wireless Info. Networks*, 19 pages, 2003.
5. Chengshan Xiao, **Jingxian Wu**, Sang-Yick Leong, Yahong R. Zheng and Khaled B. Letaief, "A Discrete-time model for spatio-temporally correlated MIMO WSSUS multipath channels", accepted for publication by *IEEE Trans. Wireless Commun.*, 28 pages, 2003.

Conference Papers

6. **Jingxian Wu**, Henry Horng, Jinyun Zhang, Andreas F. Molisch, and Chengshan Xiao, "Combining Orthogonal Space Time Block Codes with Adaptive Sub-group Antenna Encoding", submitted to *IEEE Inern. Symposium Personal, Indoor and Mobile Radio Commun. PIMRC'04*, 2004.
7. **Jingxian Wu**, Henry Horng, Jinyun Zhang, Andreas F. Molisch, and Chengshan Xiao, "Adaptive transmit diversity with quadrant phase constraining feedback", submitted to *IEEE Veh. Techno. Conf. VTC'04 Fall*, 2004.
8. **Jingxian Wu**, Chengshan Xiao, and Norman C. Beaulieu, "Error Performances of Optimum Coherent Diversity Receivers with Channel Estimation Error in Rayleigh and Ricean Fading Channels", to appear in *Proc IEEE Intern. Conf. Commun ICC'04*, 5 pages, 2004.
9. **Jingxian Wu**, Chengshan Xiao, and Khaled B. Letaief, "MMSE-Based Channel Estimation for CDMA Systems over Frequency-Selective Fading Channels", in *Proc. IEEE Veh. Techno. Conf. VTC'03 Fall*, 2003.
10. Sang-Yick Leong, **Jingxian Wu**, Jan C. Olivier, and Chengshan Xiao, "Fast Time-varying Dispersive Channel Estimation and Equalization for 8-PSK Cellular System", in *Proc. IEEE Global Commun. Conf. GLOBECOM'03 Fall*, 2003.
11. **Jingxian Wu**, Chengshan Xiao, and Jan C. Olivier, "Time-Varying and frequency-selective channel estimation with unequally spaced pilot symbols", in *Porc. IEEE Intern. Conf. Acoustic Speech & Signal Processing ICASSP'03*, Apr. 2003.
12. Chengshan Xiao, **Jingxian Wu**, and Sang-Yick Leong, "A Discrete-time model for spatio-temporally correlated MIMO WSSUS multipath channels", in *Proc. IEEE Wireless Commun. Network Conf. WCNC'03*, Mar. 2003.
13. **Jingxian Wu**, Mingquan Lu, and Zhenming Feng, "Extended ns for HFC MAC Layer Simulation", in *Proc. Intern. Conf. Telecommun. ICT'02*, Jun. 2002.

Patent

14. Jingxian Wu, Henry Horng, Jinyun Zhang, and Andreas F. Molisch, "Adaptive transmit diversity with quadrant phase constraining feedback ", patent filed by Mitsubishi Electric Research Lab.

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